AP CALCULUS AB

Unit 1 • Lesson 1.7 practice

Lesson 1.7 • Selecting Procedures for Determining Limits

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 3} \frac{x+1}{(x-3)^2}$ is

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 4

Question 2

NO CALCULATOR

$\lim_{x \to 0} \frac{x^2+5x}{x}$ is

(A) 0

(B) 5

(C) 1

(D) It does not exist.

Question 3

NO CALCULATOR

$\lim_{x \to 1} \frac{x^2+1}{x+1}$ is

(A) 0

(B) 2

(C) 1

(D) It does not exist.

Question 4

NO CALCULATOR

$\lim_{x \to 4} \frac{|x-4|}{x-4}$ is

(A) 1

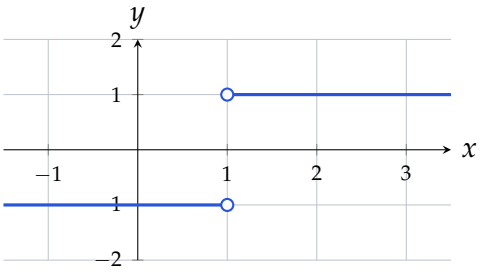
(B) -1

(C) 0

(D) It does not exist.

Question 5 (graph) NO CALCULATOR

The graph of $f(x) = \dfrac{|x-1|}{x-1}$ is shown. What is $\lim\limits_{x \to 1} f(x)$?



(A) 1
(B) -1
(C) 0
(D) It does not exist.

Question 6 (Free response) NO CALCULATOR

For each limit, name the method you choose and evaluate the limit.

(a) $\lim\limits_{x \to -1} \dfrac{x^2 + 2x + 1}{x + 1}$ (b) $\lim\limits_{x \to 2^+} \dfrac{x}{2 - x}$

(c) $\lim\limits_{x \to 9} \dfrac{x - 9}{\sqrt{x} - 3}$ (d) $\lim\limits_{x \to 0} \dfrac{\sin 2x}{\sin 3x}$